

**Eighth Semester B. E. (Electronics and Communication)
Examination**

(Elective – VI)

ANTENNA THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) All questions carry marks as indicated.
- (4) Draw neat diagrams wherever necessary.

1.
 - (a) Explain with diagrams the formation and detachment of Electric field lines for short dipole. 5(CO1)
 - (b) An antenna has field pattern given by $E(\theta) = \cos(\theta) \cos(2\theta)$ for $0 \leq \theta \leq 90^\circ$. Find :
 - (a) Half Power Beam-Width (HPBW).
 - (b) Beam-width between first nulls (FNBW). 5(CO4)
2.
 - (a) Explain the Babinet's principle for field equivalence with suitable example. 5(CO1)
 - (b) Compare front fed parabolic reflector with Cassegrain fed parabolic reflector with suitable diagrams. 5(CO1,4)
3.
 - (a) Explain with diagram the working of Yagi–Uda antenna. State its advantages and disadvantages. 5(CO4)
 - (b) Suggest the suitable feed for micro strip patch antenna in order to reduce spurious radiations. Explain such antenna with diagram. 5(CO4)

4. (a) What is meant by uniform linear array ? Explain Broadside array with diagrams. 5(CO5)
- (b) For end fire array consisting of several half wave length long isotropic radiators to have a directive gain of 30. Find the array length and width of the major lobe. What will be these values for a broadside array ? 5(CO4,5)
5. (a) Explain Switched beam systems and Adaptive array systems for Smart antennas. 5(CO2,3)
- (b) Explain the benefits of Smart antennas over conventional antennas. 5(CO2,3)
6. (a) Explain the architecture of Smart antenna system in detail. 5(CO3)
- (b) State the applications of Smart antenna system. 5(CO3)

